

F1a

Algebraic Expressions

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

(1) $2(a + 3b) =$

(2) $5(4x + 7) =$

(3) $2(4x - 3) =$

(4) $4(2y + 1) =$

(5) $4(3a + 1x) =$

(6) $5(a + 2y) =$

F2a

Algebraic Expressions

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

(1) $4(b - 2) =$

(2) $-2(x + 1a) =$

(3) $3(2y + 1) =$

(4) $3(a + 6) =$

(5) $5(2x + 3y) =$

(6) $4(y - 4) =$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

(1) $2(b + 2x) =$

(2) $-2(x + 1y) =$

(3) $-4(3a - 4b) =$

(4) $-5(x - 1a) =$

(5) $5(a - 6) =$

(6) $5(y + 4) =$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

(1) $5(2x + 2) =$

(2) $2(3y - 4b) =$

(3) $4(2a + 2) =$

(4) $2(3a - 1) =$

(5) $5(2x + 5) =$

(6) $2(x - 1b) =$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 4(x - 3a) =$$

$$(2) \quad 2(3b + 8) =$$

$$(3) \quad 4(4b + 4) =$$

$$(4) \quad 4(2a + 4) =$$

$$(5) \quad 3(3a + 2) =$$

$$(6) \quad 4(y - 3a) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 4x + 4(4x + 1y) =$$

$$(2) \quad 4(x + 2) + 3(x - 5) =$$

$$(3) \quad 4(y + 3) + 3(y - 5) =$$

$$(4) \quad 2(y + 4) + 4(y + 1) =$$

$$(5) \quad 4(a + 4) + 4(a + 4) =$$

$$(6) \quad 3(x + 1) + 4(x - 1) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 4(b + 2) + 4(b + 2) =$$

$$(2) \quad 3a + 4(4a + 1y) =$$

$$(3) \quad 4y + 3(-1y + 2x) =$$

$$(4) \quad 4(x + 4) + 4(x - 4) =$$

$$(5) \quad 4(x + 4) + 2(x - 2) =$$

$$(6) \quad 2a + 2(3a + 3y) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 4a + 4(-1a + 4b) =$$

$$(2) \quad 2y + 3(-1y + 2b) =$$

$$(3) \quad 2y + 3(-3y + 2x) =$$

$$(4) \quad 3(b + 1) + 4(b + 3) =$$

$$(5) \quad 4(a + 1) + 2(a - 1) =$$

$$(6) \quad 4(a + 2) + 3(a + 5) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 3(y + 2) + 2(y + 3) =$$

$$(2) \quad 5(a + 4) + 4(a + 3) =$$

$$(3) \quad 4(a + 4) + 3(a + 1) =$$

$$(4) \quad 2(x + 3) + 2(x + 2) =$$

$$(5) \quad 4y + 3(-1y + 1a) =$$

$$(6) \quad 3(b + 4) + 2(b - 5) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 9b^3y^2 \times 9 =$$

$$(2) \quad 4z \times (-6z) =$$

$$(3) \quad 4y^2b^2 \times 4y^2b =$$

$$(4) \quad -6x^3z \times 4x =$$

$$(5) \quad -9y \times 9 =$$

$$(6) \quad 9b \times (-8x) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

(1) $6ya \times 9ya^2 =$

(2) $5a^3y^2 \times 2y^2 =$

(3) $8z^2b \times (-6zb^2) =$

(4) $-2 \times (-7x^2) =$

(5) $2a^3 \times 5y^2 =$

(6) $9b^3x \times 5bx =$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

(1) $5a + 3(2a + 1b) =$

(2) $3y + 3(3y + 1a) =$

(3) $5(x + 1) + 2(x - 4) =$

(4) $4y + 3(3y + 4x) =$

(5) $5(x + 1) + 4(x + 1) =$

(6) $3(a + 4) + 4(a + 5) =$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

(1) $4x^2 \times 6xy^2 =$

(2) $8b^2 \times 8a =$

(3) $-9y^3 \times (-5) =$

(4) $6a^3y \times (-7ay) =$

(5) $-5b^2y^2 \times (-9y) =$

(6) $4z \times 8y^2 =$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

(1) $2x^2z \times 5x^2z =$

(2) $7x^3y^2 \times (-8xy^2) =$

(3) $4b^3y^2 \times 4y^2 =$

(4) $5a^3z^2 \times 5z =$

(5) $-9a^2x \times 9a^2 =$

(6) $6a^2x^2 \times 2a^2x^2 =$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad (y + 6)(y + 6) =$$

$$(2) \quad 4b(3b - 4) =$$

$$(3) \quad (x - 4)(x + 2) =$$

$$(4) \quad 4b(1b + 1) =$$

$$(5) \quad 3a(2a + 6) =$$

$$(6) \quad 3y(1y - 3x) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 9b^3x^2 \times 6 =$$

$$(2) \quad 7z \times (-6z^2x^2) =$$

$$(3) \quad 2a^2 \times (-4a^2) =$$

$$(4) \quad -4 \times 8y^2a^2 =$$

$$(5) \quad 4y^2 \times 4y^2b^2 =$$

$$(6) \quad 2x^3z \times (-2x^2z) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 3b(3b + 6x) =$$

$$(2) \quad 5b(3b - 2y) =$$

$$(3) \quad 4x(1x - 3) =$$

$$(4) \quad 2x(3x + 3b) =$$

$$(5) \quad (x + 3)(x - 5) =$$

$$(6) \quad 4x(2x - 3a) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 4y(3y + 6) =$$

$$(2) \quad 4b(2b + 3y) =$$

$$(3) \quad (y - 3)(y + 1) =$$

$$(4) \quad 4b(3b - 4y) =$$

$$(5) \quad (y - 6)(y - 5) =$$

$$(6) \quad 2y(3y + 4x) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

(1) $(y - 1)(y - 2) =$

(2) $(a + 2)(a + 5) =$

(3) $3a(1a - 1) =$

(4) $3a(2a + 6) =$

(5) $5x(3x - 3) =$

(6) $(a - 3)(a + 5) =$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

(1) $3y(4y - 4a) =$

(2) $3x(2x + 4) =$

(3) $(x + 4)(x + 6) =$

(4) $2y(1y + 3) =$

(5) $(a - 2)(a + 5) =$

(6) $(y + 6)(y + 3) =$

Section 1 - Solutions

10a-1:	$12y^2 + 24y$
10a-2:	$8b^2 + 12by$
10a-3:	$y^2 - 2y - 3$
10a-4:	$12b^2 - 16by$
10a-5:	$y^2 - 11y + 30$
10a-6:	$6y^2 + 8yx$
10b-1:	$y^2 - 3y + 2$
10b-2:	$a^2 + 7a + 10$
10b-3:	$3a^2 - 3a$
10b-4:	$6a^2 + 18a$
10b-5:	$15x^2 - 15x$
10b-6:	$a^2 + 2a - 15$
1a-1:	$2a + 6b$
1a-2:	$20x + 35$
1a-3:	$8x - 6$

Section 1 - Solutions (continued)

2b-1:	$2b + 4x$
2b-2:	$-2x - 2y$
2b-3:	$-12a + 16b$
2b-4:	$-5x + 5a$
2b-5:	$5a - 30$
2b-6:	$5y + 20$
3a-1:	$4x - 12a$
3a-2:	$6b + 16$
3a-3:	$16b + 16$
3a-4:	$8a + 16$
3a-5:	$9a + 6$
3a-6:	$4y - 12a$
3b-1:	$16b$
3b-2:	$-y + 6b$
3b-3:	$-7y + 6x$

Section 1 - Solutions (continued)

$$3b-4: \quad 7b + 15$$

$$3b-5: \quad 6a + 2$$

$$3b-6: \quad 7a + 23$$

$$4a-1: \quad 20x + 4y$$

$$4a-2: \quad 7x - 7$$

$$4a-3: \quad 7y - 3$$

$$4a-4: \quad 6y + 12$$

$$4a-5: \quad 8a + 32$$

$$4a-6: \quad 7x - 1$$

$$4b-1: \quad 8b + 16$$

$$4b-2: \quad 19a + 4y$$

$$4b-3: \quad y + 6x$$

$$4b-4: \quad 8x$$

$$4b-5: \quad 6x + 12$$

$$4b-6: \quad 8a + 6y$$

Section 1 - Solutions (continued)

$$1a-4: \quad 8y + 4$$

$$1a-5: \quad 12a + 4x$$

$$1a-6: \quad 5a + 10y$$

$$1b-1: \quad 10x + 10$$

$$1b-2: \quad 6y - 8b$$

$$1b-3: \quad 8a + 8$$

$$1b-4: \quad 6a - 2$$

$$1b-5: \quad 10x + 25$$

$$1b-6: \quad 2x - 2b$$

$$2a-1: \quad 4b - 8$$

$$2a-2: \quad -2x - 2a$$

$$2a-3: \quad 6y + 3$$

$$2a-4: \quad 3a + 18$$

$$2a-5: \quad 10x + 15y$$

$$2a-6: \quad 4y - 16$$

Section 1 - Solutions (continued)

5a-1:	$5y + 12$
5a-2:	$9a + 32$
5a-3:	$7a + 19$
5a-4:	$4x + 10$
5a-5:	$y + 3a$
5a-6:	$5b + 2$
5b-1:	$11a + 3b$
5b-2:	$12y + 3a$
5b-3:	$7x - 3$
5b-4:	$13y + 12x$
5b-5:	$9x + 9$
5b-6:	$7a + 32$
6a-1:	$81b^3y^2$
6a-2:	$-24z^2$
6a-3:	$16y^4b^3$

Section 1 - Solutions (continued)

7b-1:	$54b^3x^2$
7b-2:	$-42z^3x^2$
7b-3:	$-8a^4$
7b-4:	$-32y^2a^2$
7b-5:	$16y^4b^2$
7b-6:	$-4x^5z^2$
8a-1:	$10x^4z^2$
8a-2:	$-56x^4y^4$
8a-3:	$16b^3y^4$
8a-4:	$25a^3z^3$
8a-5:	$-81a^4x$
8a-6:	$12a^4x^4$
8b-1:	$y^2 + 12y + 36$
8b-2:	$12b^2 - 16b$
8b-3:	$x^2 - 2x - 8$

Section 1 - Solutions (continued)

8b-4:	$4b^2 + 4b$
8b-5:	$6a^2 + 18a$
8b-6:	$3y^2 - 9yx$
9a-1:	$9b^2 + 18bx$
9a-2:	$15b^2 - 10by$
9a-3:	$4x^2 - 12x$
9a-4:	$6x^2 + 6xb$
9a-5:	$x^2 - 2x - 15$
9a-6:	$8x^2 - 12xa$
9b-1:	$12y^2 - 12ya$
9b-2:	$6x^2 + 12x$
9b-3:	$x^2 + 10x + 24$
9b-4:	$2y^2 + 6y$
9b-5:	$a^2 + 3a - 10$
9b-6:	$y^2 + 9y + 18$

Section 1 - Solutions (continued)

6a-4:	$-24x^4z$
6a-5:	$-81y$
6a-6:	$-72xb$
6b-1:	$54y^2a^3$
6b-2:	$10a^3y^4$
6b-3:	$-48z^3b^3$
6b-4:	$14x^2$
6b-5:	$10a^3y^2$
6b-6:	$45b^4x^2$
7a-1:	$24x^3y^2$
7a-2:	$64ab^2$
7a-3:	$45y^3$
7a-4:	$-42a^4y^2$
7a-5:	$45b^2y^3$
7a-6:	$32y^2z$